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Update – 1-12th No class

25 → B.S on ans

26 → DP end

27 → Backtracking-1

30 Tue → B-2

Greedy

infix, prefix

Quick Sort

Prefix Sum

Graph

Seive

Weekend Batch Free

1) LL
2) Stack + infix
3) Queue
4) Prefix Sum
5) Seive
6) Greedy
7) BT
8) BST
9) Heap
10) Hashmap
11) Graph
12) DP
13) Back
14) B.S. on ans

Binary Search on Answer

Merge → Weekday & Weekend

Certificate – Schmeet Contest

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Ques: Koko eating Bananas

Koko

Bandariya

5 10 3 (K = 4 hours available) *3 hrs to lagege

Speed = 10 per hour 18 hrs

What we want → Koko's per hour eating speed to be minimised

lo = ~~1~~ 2 3 4 5

hi = ~~10~~ 4

m = ~~8~~ 2 3 4

$\boxed{s = 5}$ 1 + 2 + 1 4 hrs ✓

s = 2 3 + 5 + 2 10 hrs ✗

s = 3 2 + 4 + 1 7 hr ✗

s = 4 2 + 3 + 1 6 hr ✗

[1 ≤ Speed ≤ 10 ↓ max of arr]

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Ques: Smallest Divisor

arr = {1, 2, 5, 9} threshold = 6

{3, 5, 6} . t = 14

d=1 3 + 5 + 6 = 14

divisor choose karna hai

1 ≤ divisor ≤ 9
↓
max of array

d = 2 0.5 1 2.5 4.5

1 + 1 + 3 + 5 = 10 > 6 ✗

(Code is Homework)

d = 9 1 + 1 + 1 + 1 = 4 < 6 ✓

d = 17 1 + 1 + 1 + 1 = 4

d ≥ 9 → 4

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Ques: Capacity To Ship Packages Within D Days

A to B & B to A takes one day

1 trip takes 1 day

{1, 2, 3, 4, 5, 6, 7, 8, 9, 10} days = 5

55 capacity boat → 1 day ✓

* Boat Capacity minimise $\boxed{10 \le C \le 55}$

lo hi

* Array can be unsorted
You cannot sort it

C = 10 1, 2, 3, 4 5 6 7 8 9 10

C = 11 1234 56 7 8 9 10 6 ✗

C = 12 1234 56 7 8 9 10

C = 13 1234 56 7 8 9 10

C = 14 1234 56 7 8 9 10

C = 15 12345 67 8 9 10

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Ques: Capacity To Ship Packages Within D Days

{1, 2, 3, 4, 5, 6, 7, 8, 9, 10} days = 5

lo = ~~10~~ 15
hi = ~~55~~ ~~31~~ ~~19~~ ~~16~~ 14
mid = ~~32~~ ~~20~~ ~~14~~ ~~17~~ 15

ans = ~~32~~ 20 ~~17~~ 15

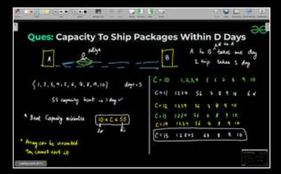
C = 32 1234567 , 8 9 10 2 days < 5

C = 20 12345 , 67 , 89 , 10 4 < 5

C = 14 1234 , 56 , 7 , 8 , 9 , 10 6 > 5

C = 17 ✓

C = 15 ✓



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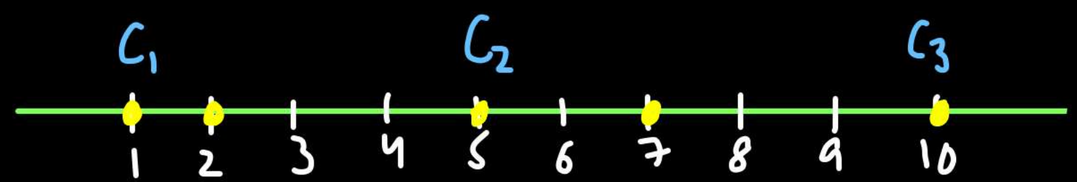
Ques: Aggressive Cows

arr = { 1, 2, 5, 7, 10 }

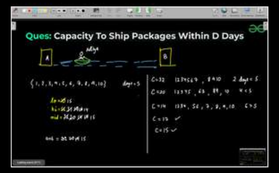
cows = 3



mindistance = 1/2

Set the cows at positions such that the minimum distance b/w 2 cows is maximum.

Maximize the min dist. b/w any 2 cows



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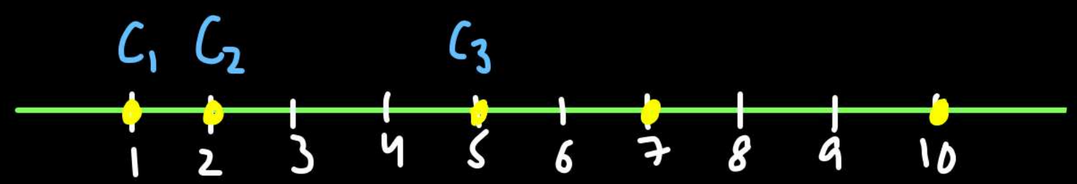
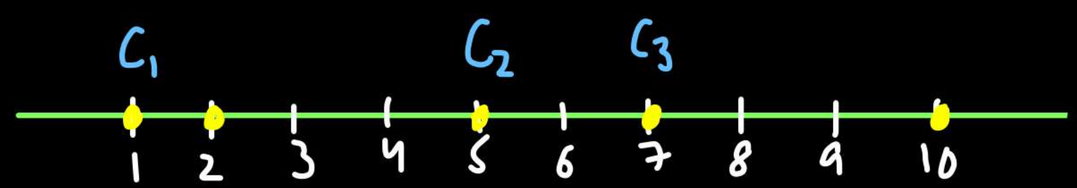
Ques: Aggressive Cows

$[1 \le \text{mindistance} \le \max(\text{le}) - \min(\text{le})]$

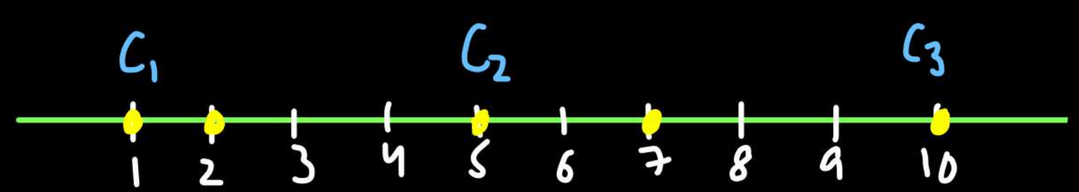
arr = { 1, 2, 5, 7, 10 } cows = 3



mindistance = 1



mindistance = 2



mindistance = 4

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(minimax)

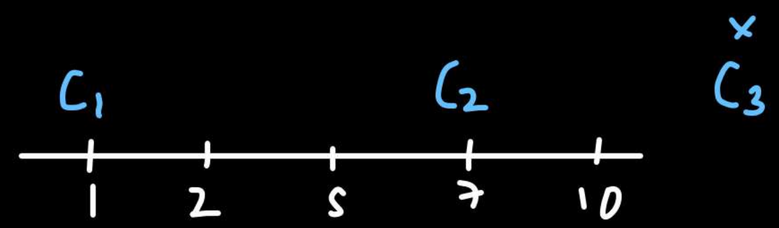
Ques: Aggressive Cows

[1 ≤ mindistance ≤ max(le) - min(le)]

$1 \le md \le 9$

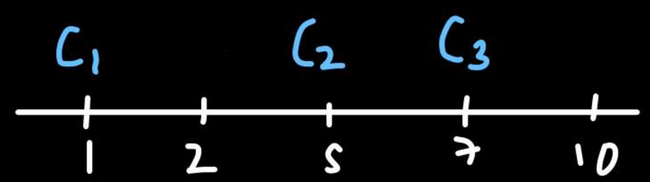
arr = { 1, 2, 5, 7, 10 }

cows = 3

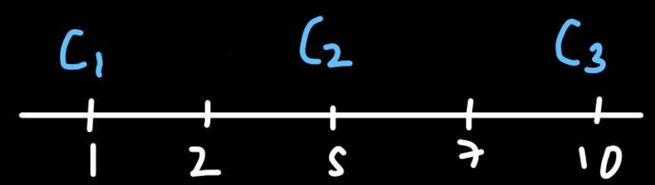


mindist = 5

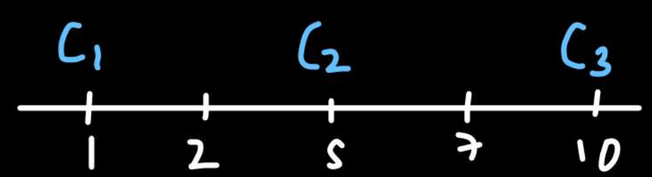
lo = ~~1~~ ~~2~~ 4 5
hi = ~~9~~ 4
m = ~~5~~ ~~7~~ ~~8~~ 4



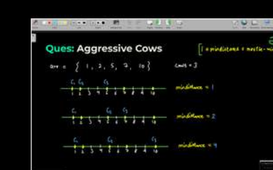
mindist = 2 ✓



mindist = 3 ✓



mindist = 4 ✓



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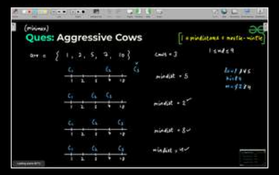
Homework: Book Allocation



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